İstanbul Kültür Üniversity

SEN0401: Special Topics in SWE Block Chain

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Content

1. Introduction (Chapter 1)
2. Overview (Chapter 2)
3. Bitcoin Core (Chapter 3)
4. Keys and Addresses (Chapter 4)
5. Wallets (Chapter 5)
6. Transactions (Chapter 6)
7. Authorization and Authentication (Chapter 7)
8. Signatures (Chapter 8)
9. Transaction Fees (Chapter 9)
10. Bitcoin Network (Chapter 10)
11. Bitcoin Blockchain (Chapter 11)
12. Mining (Chapter 12)
13. Security (Chapter 13)
14. Applications (Chapter 14)

Grading

Grading Item	BONUS	Midterm	Project	Final	Totals
%	10	30	30	40	110

Notice: This term theme is semantic technologies, so you are expected to develop industrial or educational applications for real usages, not just hypothetical. So, you need to communicate with the real users, properly experiment domain engineering facilities and collect business requirements. Team projects with well-defined responsibilities are in favour. Within that theme, the subject is Block Chain. This can be applications using available Block Chain technologies or building toy Block Chain systems or libraries for business or education! Project can also be about gamification of Block Chain teaching or using Block Chains in gamifications as well.

Textbooks

[1] Mastering Bitcoin: Programming the Open Blockchain, 3e; Andreas M. Antonopoulos & David A. Harding; O'Reilly, third edition, December 2023. Free under Creative Commons Attribution-ShareAlike 4.0: https://github.com/bitcoinbook/bitcoinbook (exact version used for this course: tag third_edition_print1, https://github.com/bitcoinbook/bitcoinbook/releases/tag/third_edition_print1; course renderings of that tag, in PDF and HTML, are distributed under the same CC BY-SA 4.0 licence)

Other references

[2] Python Blockchain, 2019, Tutorials Point.
[3] Blockchain For Dummies 2e, Tiana Laurence, Wiley, 2019.

[4] Blockchain Enabled Applications, Vikram Dhillon & David Metcalf, Springer, 2017.
[5] Develop a blockchain application from scratch in Python, IBM, Web Site.
[6] Bitcoin: A Peer-to-Peer Electronic Cash System, Satoshi Nakamoto, 2009.
[7] Handbook of Digital Currency, Bitcoin, Innovation, Financial Instruments, and Big Data; Ed: David Lee Kuo Chuen; Elsevier, 2015.
[8] Verifiable Credentials Data Model v2.0; W3C Recommendation, 5 May 2025. https://www.w3.org/TR/vc-data-model-2.0/
[9] Decentralized Identifiers (DIDs) v1.0; W3C Recommendation, 9 July 2022. https://www.w3.org/TR/did-core/ (v1.1 is a W3C Candidate Recommendation dated 5 March 2026: https://www.w3.org/TR/did-1.1/)
[10] W3C Blockchain Community Group. https://www.w3.org/community/blockchain/
[11] Ethereum: A Secure Decentralised Generalised Transaction Ledger (the Yellow Paper); Gavin Wood. https://ethereum.github.io/yellowpaper/paper.pdf

Course Hours

Lecture	Content	Date	Hours	Venue
Hour-1	Theory & Practices	Friday	09:00-09:45	LAB 2B-12/14
Hour-2			10:00-10:45	
Hour-3			11:00-11:45	
Hour-4			12:00-12:45	

Academic Integrity

The Computer Engineering Department of IKU draws your attention to the concepts of "**Academic Integrity**". You should **obey** to written and verbal rules of studentship, otherwise you will be reported to the faculty and **lose** all bonus fortunes of this course.